
Data Quality as a Causal Variable in Smartphone Health Monitoring

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Abstract: Smartphone-based health monitoring studies commonly treat data quality as a preprocessing step: flag or remove low-quality recordings, then proceed with the remainder. This places data quality outside the model when it belongs inside, as part of the data-generating process – if it is associated with the outcome of interest, filtering on it changes what the remaining data represent. We test this on the mPower 2.0 Parkinson’s disease dataset (81 participants, 15,398 resting tremor recordings), applying an automated assessment framework at three levels – study-level session completion, task-level recording duration, and signal-level flat-segment detection – and testing each check against diagnosis at the participant level. The three behave differently. Duration filtering removes recordings from Parkinson’s disease participants (clustered odds ratio 2.23, 95% CI 1.50–3.32); flat-signal filtering removes control recordings (OR 0.52, 0.23–1.19), because on a resting tremor task low signal variation is the expected observation for a participant without tremor rather than a measurement failure; session completion shows no association once follow-up length is accounted for (OR 1.13, 0.53–2.37). Filtering on data quality is therefore not neutral, and its direction cannot be assumed. The main classification results are nonetheless stable after duration filtering. Data quality belongs inside the causal model as separate variables, each tested against the outcome before it is filtered on.

Keywords: data quality, protocol adherence, causal inference, smartphone health monitoring, Parkinson’s disease, directed acyclic graphs

1. Introduction

Smartphone-based health monitoring produces large numbers of repeated recordings collected without clinical supervision. Because participants complete tasks in their own homes, the recordings vary in quality. Some are too short or too long. Some show very little signal variation, which is usually taken to indicate the device was left still during the task. Some participants do not complete enough sessions over time. These problems are well documented [1–4], and the standard response in the literature is to filter: define quality criteria, remove recordings that fail them, and proceed with the remainder [5–7].

This approach rests on a hidden assumption: that filtering on data quality is a neutral step that removes noise without introducing bias. This paper argues the assumption does not hold in general. If the data quality variable is associated with the outcome of interest – which is easy to check but rarely checked – then conditioning on it changes the composition of the sample in ways that may distort later analysis. In causal terms, a variable that lies between other variables of interest should be inside the model, not removed from it before analysis begins [8–10].

The correct response is to represent data quality inside the causal model, where its relationship to other

variables can be examined and reported rather than left invisible.

This paper makes two contributions. First, we present a structured automated assessment framework that operates at three levels of the data-collection process. Second, we provide empirical evidence, using a real Parkinson’s disease dataset, that quality filtering is not neutral, and that the direction in which it is not neutral differs by check: one check removes recordings over-represented among PD participants, one removes recordings over-represented among controls, and one removes neither. This has a consequence for how data quality should be represented in a causal model, which we set out alongside what it means for how health monitoring pipelines should be built and reported.

2. Background

2.1. Data quality in remote smartphone health studies

Smartphone-based Parkinson’s disease monitoring has advanced substantially over the past decade, and features derived from accelerometer signals can distinguish PD participants from controls and track symptom change over time [1, 3, 4, 11]. The mPower study [11] established that remote sensing is feasible at scale, and subsequent analyses showed that careful quality control and model evaluation are necessary for reliable estimates [3, 12]. Data collected without supervision nonetheless vary considerably in quality, and participant engagement varies over time in ways that are not random [1, 2].

Automated methods for quality control have been developed across environmental and health sensing domains to detect anomalous or invalid signals before analysis [5, 13]. In Parkinson’s disease research specifically, Badawy et al. [6] showed that automated quality control for sensor-based symptom measurement outside the lab improves the reliability of extracted features. Similar approaches have been applied to voice-based monitoring [14], and walking test pipelines have demonstrated the need for structured quality checking before clinical measures can be derived [7]. These methods share a common structure: identify low-quality recordings, remove them, and treat the remainder as a valid dataset. The relationship between the quality filter and the outcome variable is not examined.

2.2. Data quality as a causal problem

In causal inference the data-generating process is represented as a directed acyclic graph (DAG), in which directed edges represent causal relationships and the graph determines which conditioning operations are valid [8].

Selection bias arises when the analysed sample is systematically different from the target population, and when the selection mechanism is related to variables in the model [9]. Data quality filtering is a form of selection. If the quality variable Q is influenced by the outcome or exposure of interest, conditioning on Q selects a non-representative subset and can distort the relationships the analysis is trying to estimate. This problem is not resolved by using better classifiers or larger samples – it is a structural problem in the analytical pipeline [10, 15].

Representing Q as a node in the DAG makes this risk explicit. The filtered dataset no longer represents the same population as the unfiltered one, and this should be stated.

3. Causal Framework and Quality Assessment

3.1. Causal model

The data-generating process for smartphone accelerometer recordings in mPower 2.0 is represented using a DAG with eight nodes: gender (G), age, medication state (M), time of day (T), Parkinson’s symptoms (P), accelerometer data (Acc), data quality (Q), and individual change (N). Figure 1 shows the structural relationships.

Data quality Q is placed inside the DAG because poor-quality or non-adherent recordings distort how symptom-related movement is represented in the accelerometer signal [1, 6]. Q affects Acc , and symptoms P are drawn as a cause of Q , so Q is a mediator between P and Acc . More severe symptoms can change

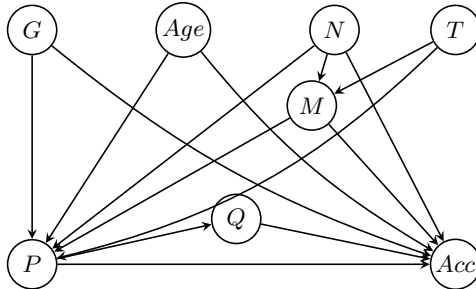


Figure 1. DAG for smartphone-based Parkinson’s disease monitoring. Eight nodes: gender (G), age, time of day (T), individual change (N), medication state (M), symptoms (P), data quality (Q), and accelerometer data (Acc). Q is inside the model as a node that affects Acc , and symptoms P are drawn as a cause of Q ($P \rightarrow Q$), making Q a mediator between P and Acc . Section 4.2 provides evidence consistent with this edge for the duration check, but not for the other two components of Q . Section 5 discusses why the single node shown here is not supportable.

how a participant performs the task, which affects the quality of the recording. The empirical results in Section 4.2 support this edge for one of the three quality checks, do not support it for a second, and indicate a relationship in the opposite direction for the third. Section 5 returns to what this means for the single node. The structural point is that Q is inside the model. Treating it as a preprocessing step removes it and makes its relationships to other variables unexaminable.

3.2. Automated quality assessment framework

An automated framework was applied to mPower 2.0 recordings to characterise data quality at three levels, shown in Table 1, following the structure proposed in [5, 6, 14].

Table 1. Automated quality assessment framework applied to mPower 2.0.

Check	Level	Criterion
Session completion	Study	<10 completed days per quarter
Duration adherence	Task	Duration <28s or >32s
Flat signal detection	Signal	>10% of 1s windows below the 5th percentile of pooled rolling SD

Session completion monitoring assesses study-level adherence. Participants were expected to complete sessions on at least 10 days per quarter [3]. Figure 2 shows an example of an adherent and a non-adherent participant. Quarters with fewer completed days were flagged as non-adherent. When participants do not meet this requirement, the resulting data may contain important gaps in longitudinal coverage, reducing the reliability of repeated measurements [2].

Task duration adherence assesses whether each recording matches the expected task duration. The resting tremor task was designed to last 30 seconds. Duration was computed from the difference between the first and last timestamps in each sensor file [7]. Recordings shorter than 28 seconds or longer than 32 seconds were flagged. The excluded recordings are mostly short, with mean 21.95s and median 23.53s against a minimum of 3.28s; a small number run far beyond the intended duration, to a maximum of 496s.

Flat signal detection assesses whether the accelerometer signal contains sufficient movement variation to be interpretable as a valid motor task recording. A rolling standard deviation was computed over one-second windows of the dynamic acceleration magnitude, following the general approach to signal-level quality control in [6]. A recording was flagged when more than 10% of its windows fell below a threshold set at a percentile of the pooled rolling-standard-deviation distribution across all recordings. Because this percentile is not fixed by the protocol, Section 4.2 reports results across the 5th, 10th and 20th percentiles rather than at a single value. Figure 3 shows a flat and a non-flat tremor signal. The check treats the flat recording

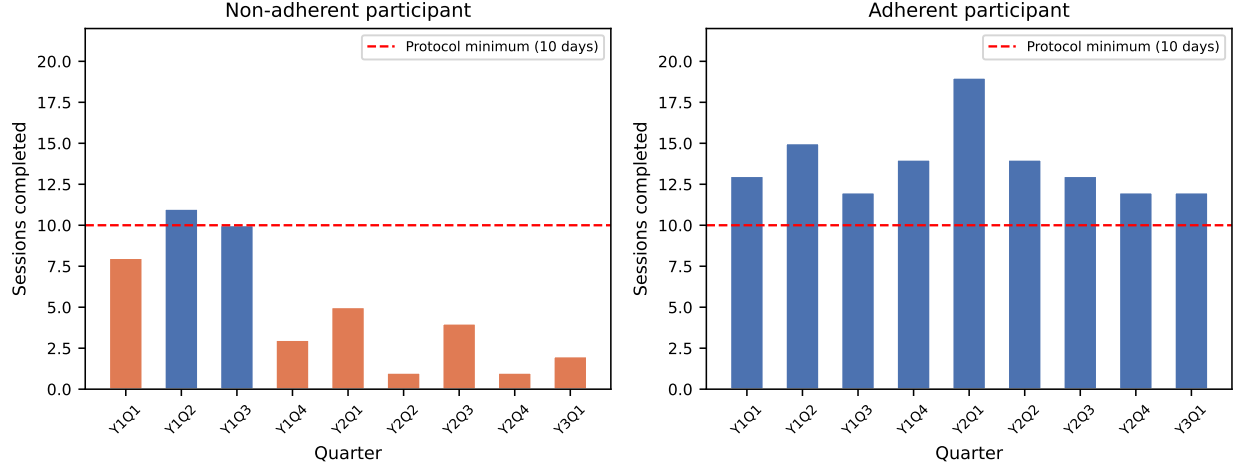


Figure 2. Session completion per quarter for a non-adherent participant (left) and an adherent participant (right). Red bars fall below the 10-day protocol minimum. Adherent participants complete 12–19 sessions per quarter consistently; the non-adherent participant drops to as few as 1 session in some quarters.

as a case where the participant did not sustain the task as intended. Section 4.2 examines whether that interpretation holds on this task.

The three checks address different levels of the data-collection process. A participant may satisfy one without satisfying the others: a full quarter of sessions may still contain short recordings, and a normal-duration recording may still contain a flat signal. Keeping them separate makes it possible to examine each type of protocol departure independently [6, 15]. Their combined outcome is summarised in the DAG by the single node Q . Section 4.2 tests each check separately against diagnosis and finds that they do not share a common relationship to it; Section 5 sets out why the single node is therefore not supportable.

4. Empirical Analysis

4.1. Dataset

The analysis uses mPower 2.0 resting tremor data [3, 11]. After preprocessing following [15], the dataset contains 15,398 tremor recordings from 81 participants: 57 with Parkinson’s disease (PD) and 24 healthy controls. Mean participant age is 69.4 years (range 40–84). The PD and control groups are closely age-matched (mean 69.6 vs. 68.9 years). In the full dataset, 67% of recordings come from PD participants and 33% from controls. The number of valid recording days per participant ranged from 2 to 200 (mean 91.4, SD 38.6).

4.2. The three quality checks relate to diagnosis differently

Each of the three checks in Table 1 was tested against diagnosis. All three were assessed at the participant level, because recordings are clustered within participants and a recording-level test would treat correlated observations as independent. Associations are reported as odds ratios from logistic generalised estimating equations (GEE) with an exchangeable working correlation, clustered by participant, following [9]. Results are summarised in Table 2.

4.2.1 Duration adherence removes PD recordings

Recordings with duration between 28 and 32 seconds were retained as the duration-adherent subset, reducing the dataset from 15,398 to 5,741 recordings while keeping all 81 participants.

Figure 4 shows the shift in composition: PD participants supply 74% of the excluded recordings ($Q = 0$, $n = 9,657$) against 67% of the full dataset, and 56% of those retained. At the participant level the median

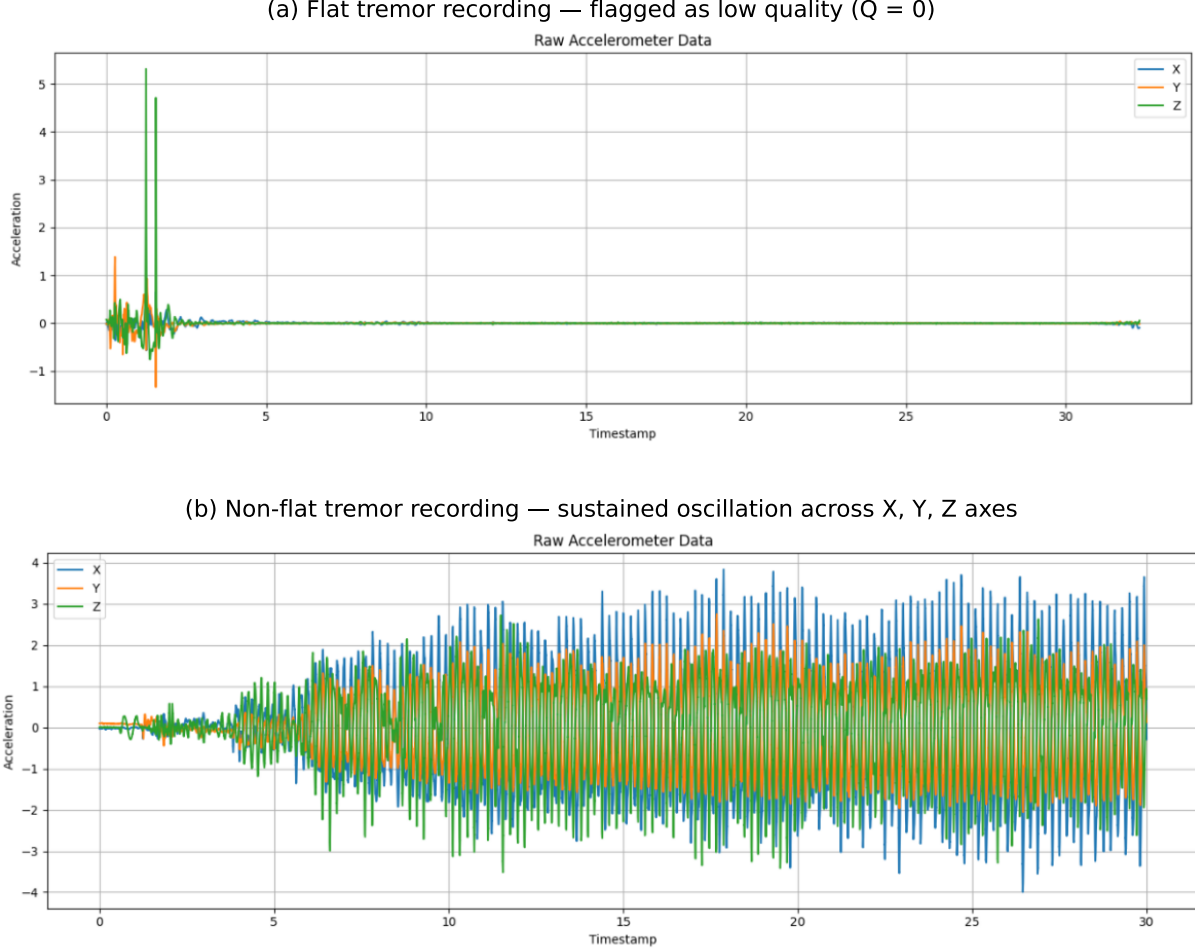


Figure 3. (a) A flat tremor signal: brief spikes at the start followed by near-zero variation across the recording. The check flags this as poor quality, on the assumption that near-zero variation indicates a device left still rather than a task performed. (b) A non-flat tremor signal: sustained oscillation at approximately 5–6 Hz, consistent with resting tremor during the intended task. Section 4.2 shows that on a resting tremor task the assumption behind (a) does not hold in general, because a participant without tremor also produces low variation.

exclusion rate was 72% for PD participants and 54% for controls (Mann-Whitney U test, $p < 0.001$), and the clustered odds ratio for exclusion given PD was 2.23 (95% CI 1.50–3.32), or 2.22 after adjusting for age.

This is consistent with symptoms influencing recording behaviour: participants with more severe tremor or motor difficulty may find it harder to sustain the task for the full 30 seconds, producing shorter recordings that fail the check [1, 4]. This is the pattern expected under the $P \rightarrow Q$ edge in Figure 1, though the association does not by itself rule out shared causes such as engagement, discussed in Section 5.

4.2.2 Session completion shows no association once follow-up is accounted for

Session completion was assessed over 609 participant-quarters. Of the 112 quarters falling below the 10-day threshold, 75% came from PD participants against 66.7% of all quarters – a shift closely resembling the duration result.

That resemblance does not survive testing. The median proportion of below-threshold quarters was 11.1% in both groups (Mann-Whitney U test, $p = 0.29$), the proportion of participants adherent in every quarter was almost identical (36.8% of PD participants, 37.5% of controls; Fisher exact $p = 1.00$), and the unadjusted clustered odds ratio was 1.71 (95% CI 0.80–3.66).

The reason is follow-up length. Adherence requires meeting the threshold in every quarter, so it is stricter

Table 2. Each quality check tested against diagnosis. Odds ratios are from participant-clustered GEE logistic regression; an OR above 1 means PD recordings are flagged more often, below 1 means control recordings are flagged more often. The three checks point in three different directions.

Check	Level	Flagged	% PD	OR (95% CI)
Duration adherence	Task	9,657	74%	2.23 (1.50–3.32)
Session completion	Study	112	75%	1.13 (0.53–2.37)
Flat signal	Signal	2,187	56%	0.52 (0.23–1.19)

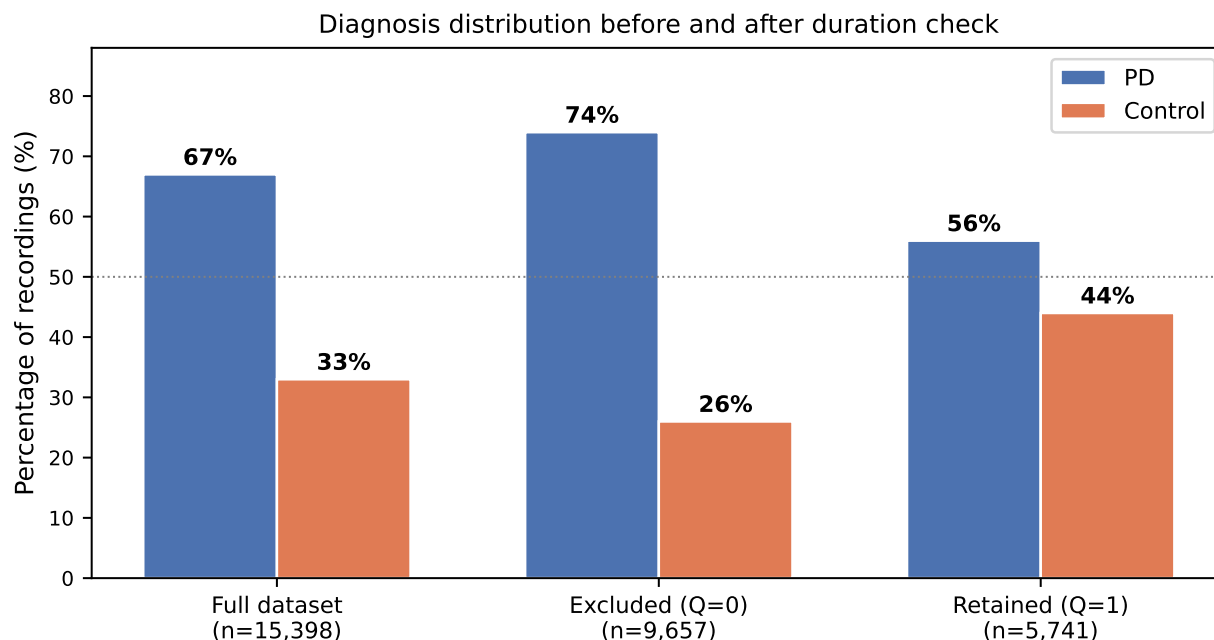


Figure 4. Diagnosis distribution across the full dataset, recordings excluded by the duration check ($Q = 0$), and recordings retained ($Q = 1$). PD participants are over-represented in excluded recordings (74%) relative to the full dataset (67%) and the retained subset (56%). The duration check is the only one of the three for which the shift is in this direction; see Table 2.

for participants observed longer, and controls in this dataset were followed for longer than PD participants (median 9 quarters against 8, $p = 0.03$). A participant’s final quarter is usually partial and therefore nearly always fails, so a shorter record mechanically inflates the proportion failing. Adjusting for the number of quarters reduces the odds ratio to 1.13 (95% CI 0.53–2.37), and further adjustment for age leaves it unchanged.

Taken alone, 75% against 66.7% would have supported the same conclusion as the duration check. It is the clearest case in these data of a recording-level proportion misleading.

4.2.3 Flat signal detection removes control recordings

The threshold for the flat signal check is a percentile of the pooled rolling-standard-deviation distribution and is not fixed by the study protocol, so results are reported at the 5th, 10th and 20th percentiles. Table 2 tabulates the 5th, as the swept value closest to the proportion flagged by the fixed per-axis threshold rule of [6]: that rule flags 1,678 recordings (10.9%) on this dataset against 2,187 (14.2%) at the 5th percentile.

The direction is opposite to the duration check. Of the recordings flagged at the 5th percentile, 56% come from PD participants, below the 67% dataset baseline; the median flat rate per participant was 1.3% for PD participants and 7.1% for controls; and the clustered odds ratio was 0.52 (95% CI 0.23–1.19), or 0.40 (0.16–1.01) after adjusting for age.

The direction is stable across the sweep and strengthens as the threshold loosens – 0.52, then 0.45 (0.23–

0.87) and 0.31 (0.18–0.56) – with the same ordering after age adjustment and within the duration-adherent subset. Age adjustment strengthens rather than weakens the association at every threshold, so age is not driving the result. The 20th percentile estimate should be read with caution, since the check there flags 41.7% of all recordings, which is not a plausible rate of quality failure; the conclusion rests on the common direction rather than any single magnitude.

The explanation is specific to the task. In a resting tremor recording a participant without tremor produces genuinely little variation, so low variation is the expected observation for a control rather than a failure of data collection. The rule therefore partly measures the outcome it is meant to be independent of, and filtering on it preferentially removes the control recordings that most clearly show the absence of tremor: it removes signal rather than noise.

4.2.4 Implications for the causal model

The three checks were introduced in Section 3.2 as components of a single data quality variable Q . They do not share a common relationship to diagnosis: the odds ratios are 2.23, 1.13 and 0.52, spanning both directions. Section 5 sets out what this means for the DAG in Figure 1 and why the single node is not supportable.

4.3. Classification results are stable after quality filtering

To examine whether the main results depend on which recordings are retained, the diagnosis prediction analysis was repeated on the duration-adherent subset. Three models were compared under both record-wise and subject-wise cross-validation, following the identity deconfounding approach in [16] and the cross-validation evaluation framework in [12, 17]:

1. **Confounded:** sensor features only, no deconfounding.
2. **Weighted (IdBin IPW):** sensor features with inverse-probability weights based on $P(\text{Diagnosis} \mid \text{IdBin})$, reducing identity confounding without using IdBin at prediction time [16].
3. **Sensor + IdBin:** sensor features plus IdBin as an input feature (diagnostic upper bound; not for deployment).

Table 3. Diagnosis prediction on the duration-adherent subset ($n = 5,741$ recordings, 81 participants). Majority-class baseline is 56%. Values are means across five cross-validation folds; PD is the positive class.

CV	Model	Acc.	Prec.	Rec.	Spec.	F1	AUC
Record-wise	Confounded	0.712	0.752	0.729	0.690	0.740	0.802
Record-wise	Weighted (IdBin IPW)	0.706	0.782	0.662	0.762	0.717	0.799
Record-wise	Sensor + IdBin	0.813	0.834	0.834	0.786	0.834	0.902
Subject-wise	Confounded	0.590	0.651	0.688	0.550	0.638	0.699
Subject-wise	Weighted (IdBin IPW)	0.605	0.679	0.627	0.647	0.627	0.700
Subject-wise	Sensor + IdBin	0.372	0.482	0.596	0.130	0.491	0.374

Table 3 shows the results, and the pattern across models matches the full-data results in [16]. The confounded record-wise accuracy (0.712) sits about 15 percentage points above the 56% majority-class baseline; the larger gap relative to the full data reflects the more balanced class distribution after filtering rather than any gain in signal.

The additional metrics show that the small accuracy difference between the confounded and weighted models conceals a larger change in behaviour. Record-wise, inverse-probability weighting raises precision from 0.752 to 0.782 and specificity from 0.690 to 0.762 while lowering recall from 0.729 to 0.662, at essentially unchanged discrimination (AUC 0.802 against 0.799). Weighting does not remove signal; it moves the operating point, trading sensitivity for specificity as the identity-driven majority is downweighted. Accuracy alone would have suggested a marginal loss where the change is a redistribution of errors.

The Sensor + IdBin model behaves as intended in its diagnostic role. Record-wise it is the strongest model on every metric (AUC 0.902), which is expected when the same participants appear in training and test folds. Under subject-wise cross-validation it collapses: accuracy 0.372, specificity 0.130, and AUC 0.374, below the 0.5 expected from random ranking. A model that is worse than chance on unseen participants has learned bin-to-label associations that invert out of sample, which is stronger evidence than a below-baseline accuracy that IdBin must never be used as a prediction feature [16, 17].

Subject-wise estimates vary considerably across folds – the standard deviation of precision exceeds 0.17 for every model – which is expected with 81 participants split five ways, and is the relevant uncertainty for generalisation to new individuals.

The main conclusion is that quality filtering does not change the structure of the results. It changes the class balance and removes recordings in a way related to diagnosis, but the analytical conclusions hold in both datasets.

5. Discussion

5.1. Filtering on Q is not neutral, and not neutral in one direction

The duration check removes recordings in a way related to diagnosis, consistent with symptom severity influencing recording behaviour, an issue raised in prior work [1, 3, 4]. If more severely affected participants terminate the task early, filtering on duration preferentially removes the most symptomatic recordings, and the direction of influence runs from P to Q . Filtering on a mediator blocks part of the causal path from P to Acc – a different and more consequential problem than removing noise [8, 9].

The other two checks show that this is not the general case. Session completion is unrelated to diagnosis once follow-up length is accounted for, and flat signal detection is related in the opposite direction. The claim that quality filtering is not neutral therefore holds; the stronger claim, that it consistently removes the most symptomatic data, does not. Each filter distorts the sample in its own direction, and that direction cannot be predicted from the fact that a check exists.

The flat signal case is structurally distinct. It is not a mediator problem but something closer to conditioning on a descendant of the outcome: the rule is written in terms of the quantity the task is designed to measure, so applying it removes the clearest negative cases. This is invisible unless the check is tested against the outcome before it is applied.

A study that applies any quality filter and then compares groups is working with a dataset whose composition the filter has changed, in a direction that must be measured rather than assumed. At minimum the outcome distribution should be compared before and after every filter; where they differ, the analysis is answering a different question and should say so.

5.2. The single-node treatment of Q is not supportable

The framework in Section 3.2 summarises three checks in a single node Q , on the grounds that all three ask whether a recording is a valid observation of the intended motor task. Section 4.2 shows that this shared purpose does not imply a shared relationship to P . Combining the three into one node would average associations running in opposite directions and report approximately no relationship between Q and diagnosis, while each component filter continues to distort the sample in its own way. The single node does not simplify the model; it conceals the effect it was introduced to represent.

Respecifying the DAG is not straightforward, because the three checks do not stand in the same structural relation to the graph: duration adherence behaves as a mediator between P and Acc ; session completion shows no association with P and may be independent of it or related through unmeasured engagement; flat signal detection is better understood as a partial measurement of P . Three quality nodes would carry three different edge structures, and identifiability would need re-deriving rather than assuming it carries over [8, 10]. We report the empirical result and leave the respecified graph to future work.

Whether a quality check is a mediator, an independent nuisance, or a proxy for the outcome is an empirical question with a different answer for each check, and it can only be asked if the checks are kept separate and tested individually against the outcome.

5.3. Implications for study design and reporting

Three concrete changes follow from these results.

First, quality flags should be stored alongside the data and included in any published dataset release, as recommended for environmental sensing [5, 13] and for digital health pipelines [6, 7].

Second, the outcome distribution should be compared before and after every quality filter, and where they differ the filtered analysis should be presented separately with a statement of what population it represents.

Third, an association between a quality check and the outcome is a finding worth reporting rather than a nuisance to be removed. The over-representation of PD participants among short recordings is informative about how symptoms affect task execution in real-world conditions [1, 3], and the over-representation of controls among low-variation recordings is informative about what the task measures.

6. Conclusions

This paper has argued that data quality is a causal variable, not a preprocessing step. Testing three quality checks against diagnosis in mPower 2.0 resting tremor data showed that they behave differently: the duration check removes recordings from PD participants, the flat signal check removes recordings from controls, and session completion removes neither once follow-up length is accounted for. Filtering on data quality changes the recording-level diagnosis distribution, and the direction of that change differs by check. It is not a neutral operation, and it is not neutral in a predictable direction.

The classification results are nonetheless unchanged: the relative ordering of confounded, identity-deconfounded and IdBin-augmented models [16] is the same in filtered and unfiltered data. The analytical conclusions are stable, but the two datasets represent different subsets of recording behaviour.

Data quality therefore belongs inside the causal model, with its components represented separately rather than as one node and each association examined and reported. Filtering is a decision about scope that affects interpretation, not a technical step preceding the analysis.

Future work should respecify the DAG with the three checks as distinct nodes and re-derive identifiability under that graph [8, 10], confirm the $P \rightarrow Q$ edge for the duration check as a direct effect separate from shared causes such as engagement, and test whether the divergence reported here holds across the walking and tapping tasks and in other datasets.

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A. Quality Check Framework Across Tasks

Table 4 shows how the three quality checks apply across the three motor tasks in mPower 2.0. Only the resting tremor task was used in the main analysis. Session completion and duration adherence transfer directly, the latter with task-specific thresholds [7]. Flat signal detection does not transfer in the same way. On the walking and tapping tasks the participant generates movement throughout, so near-zero variation indicates a device left still and the check measures what it is intended to measure. On the resting tremor task the participant is asked to hold still, so signal variation is the quantity of clinical interest rather than a precondition for it, which is why the check behaves as reported in Section 4.2. The validity of a signal-level quality rule therefore depends on whether the task expects movement. Whichever task is used, the comparison of outcome distribution before and after filtering should be performed for each check separately before analysis.

Table 4. Quality check applicability across mPower 2.0 motor tasks. Flat signal detection is the only check whose interpretation depends on the task: it is a valid quality rule where the task expects movement, and partly a measurement of the outcome where the task expects stillness.

Check	Tremor	Walking	Tapping
Session completion	Applies	Applies	Applies
Duration adherence	28–32 s	28–32 s	Task-specific
Flat signal detection	Confounded with outcome	Applies	Applies